

EMERTXE INFORMATION TECHNOLOGIES

IOT BASED INTERNSHIP

PROJECT PRESENTATION ON HOME AUTOMATION SYSTEM

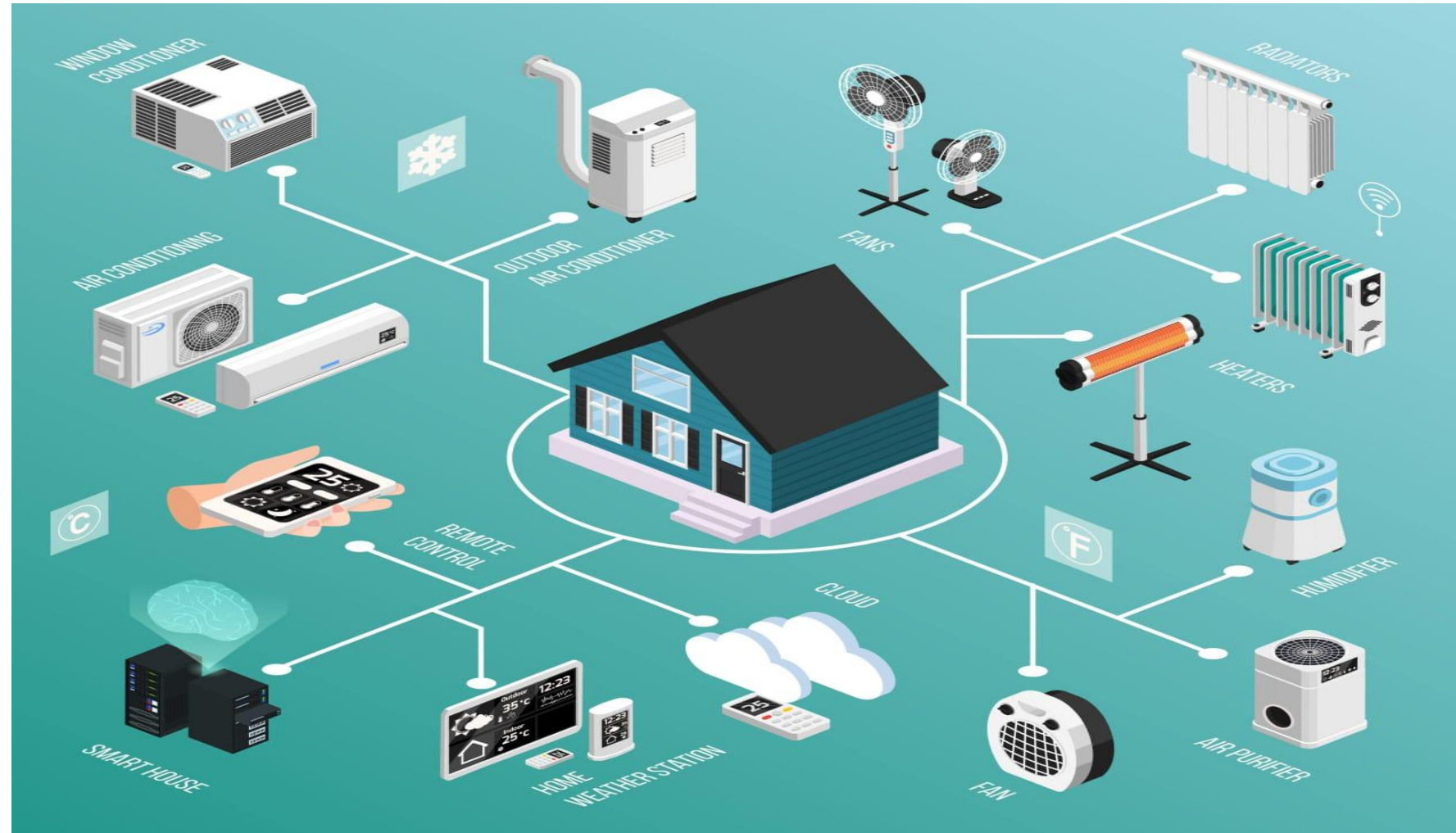
SUBMITTED BY:

G.SUNITHA

T.SRAVANTHI

CH.PARAMESWARI

M.SRUTHI



INTERNSHIP OVERVIEW

- This internship primarily aims to build understanding of the Internet of Things(IOT) and Embedded systems.
- Learn the relevance and role of IOT in everyday applications .
- Evaluate participants through weekly assessments .
- Rein force learning using simulation-based projects .
- Offer mentorship to beginners entering the IOT domain .

CONCEPTS COVERED IN THIS INTERNSHIP

- C programming : Topics include keywords , data types , operations ,loops, arrays , functions , pointers and memory usage .
- C++ programming : Focus on Object Oriented Programming concepts.
- IOT: Definitions , architectures and practical uses.
- Embedded Systems : Key peripherals such as Arduino Uno , LEDs ,CLCD , Sensors(LDR,LM35) and PWM concepts .

C PROGRAMMING

C is a widely respected , general-purpose programming language designed in the early 1970s by Dennis Ritchie at Bell Laboratories . Known for its portability , C is both powerful and efficient , forming the backbone of Operating Systems , application software and Embedded systems . Its syntax encourages structured programming , enabling developers to organize code into well-defined functions , manage memory directly , and implement complex logic using variables , constants and a variety of data types such as integers, floating -point numbers ,arrays , pointers and user defined structures . C supports procedural programming , recursion and static type checking , which helps programmers write clear , maintainable programs while maintaining high performance . The language's concise and readable syntax makes it accessible to learners , yet it is flexible enough for experienced developers to create efficient software for diverse platforms .

ABOUT C++

- It is an object-oriented language , supporting key concepts like classes , objects , encapsulation , inheritance , polymorphism and abstraction .
- C++ allows both high-level and low-level programming making it suitable for system/software development and large-scale application projects .
- The language is statically ,typed , meaning variable types are set at compile time for better performance and error detection .
- C++ programs are machine independent and portable-code can run on any platform with the appropriate compiler .
- It is compiled and provides fast execution , making it preferable for performance – critical applications .
- C++ supports features of both procedural and generic programming in addition to its object-oriented paradigm .

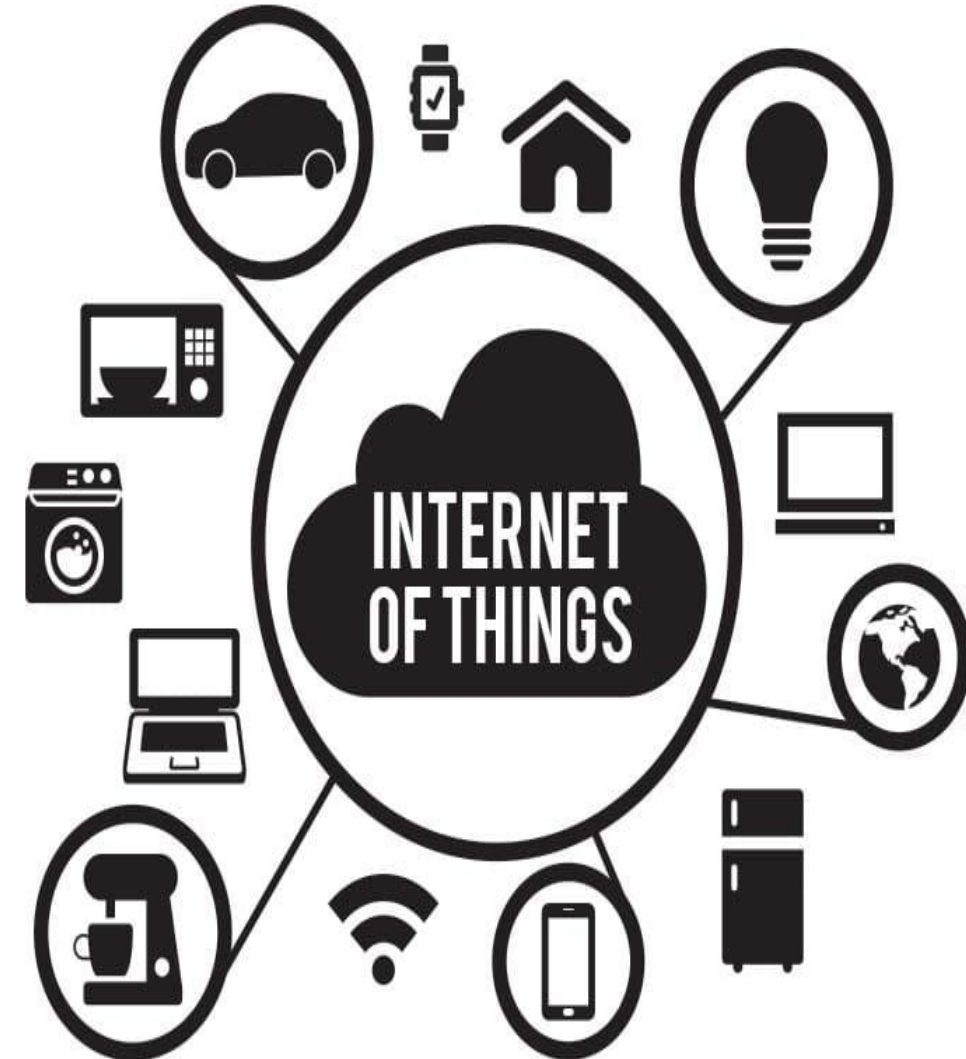
INTRODUCTION

- Key components include smart lights , thermostats , locks , cameras ,appllicants , motion sensors , actuators for controlling physical like binds or valves.
- Automation enables scheduled routines or trigger-based actions(for example , lights turning on when motion is detected or temperature adjusting based on presents an time of day)
- Security is enhanced with smart locks alarms , windows/door sensors , surveillance cameras , and instant notifications of unusual activity .
- Voice assistants(such as amazon Alexa , Google assistant) can be integrated for hands-free operation of devices or custom automation .
- Smart home systems save energy , increase convenience , enhance comfort , and improve the safety of the home environment .

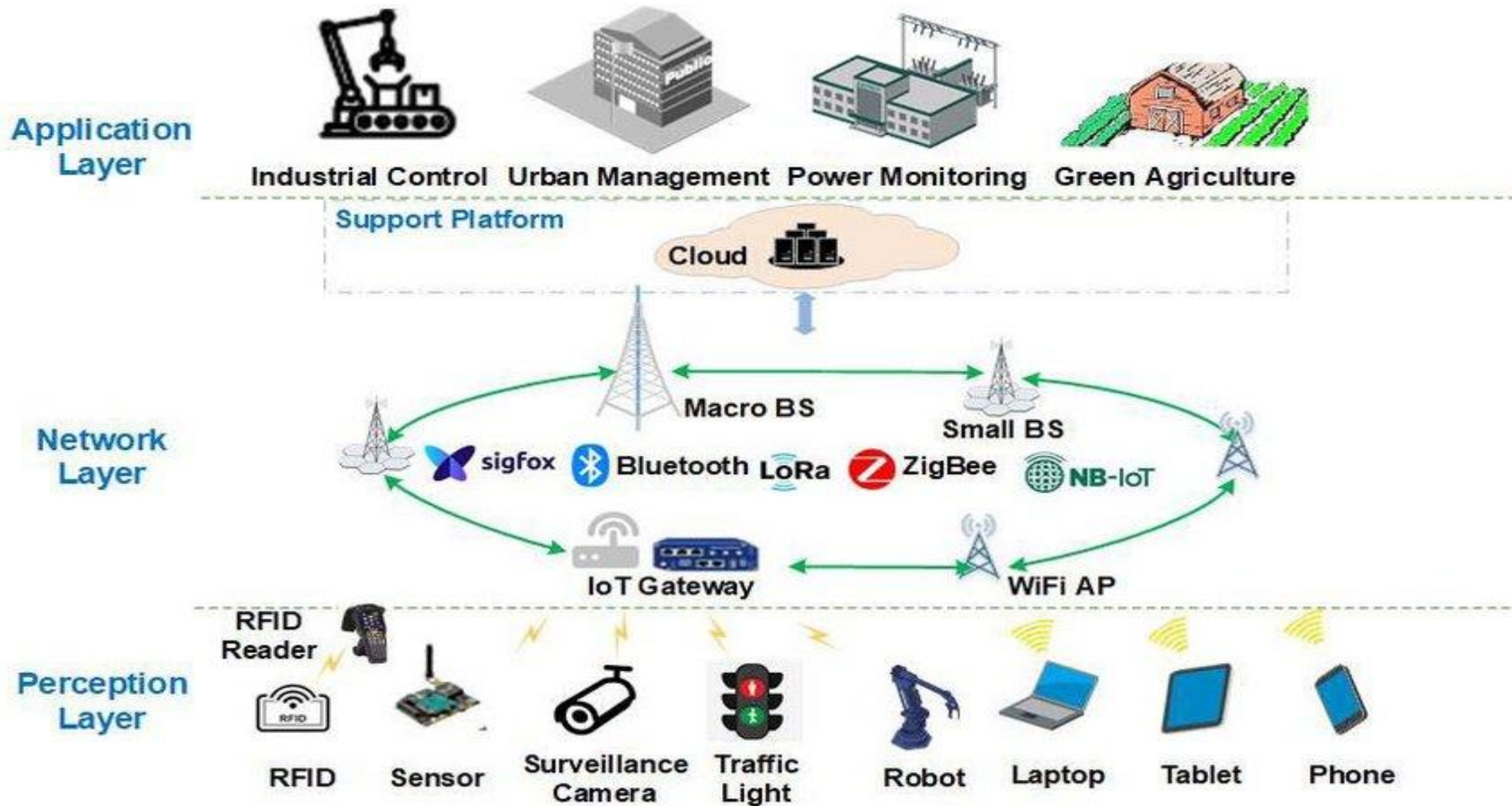


INTERNET OF THINGS

- The Internet of Things(IoT) is a network of physical objects such as devices , applicants , or sensors that are embedded with technology allowing them to connect and exchange data over the internet .
- IoT systems consists of three core elements: smart devices(sensors or actuators) , connectivity infrastructure , and data processing mechanisms(cloud or edge analytics).
- Smart devices gather environment data and send it through gateways or network protocols like Wi-fi , Bluetooth , Zigbee or cellular data , enabling interaction and remote management .
- The data collected is then analyzed (often with AI or Machine Learning algorithms) to create actionable insights , trigger automation or alert users about significant events .
- Users interact with or monitor IoT devices primarily through graphical users interfaces(mobile apps , dashboards , or web portals).



ARCHITECTURAL OVERVIEW



CHALLENGES IN IOT

- Security: IoT devices are frequent targets for cyberattacks due to their network connectivity and often weak security features , making robust protection a top priority .
- Privacy: massive data collected by IoT systems creates serious concerns around user privacy and unauthorized data access .
- Interoperability : devices from different manufacturers must communicate seamlessly ,but lack of standardization can cause compatibility issues and fragmentation .
- Scalability : managing , updating and tracking the rapidly growing number of devices in a large –scale IoT Deployment is increasingly complex .
- Connectivity: reliable and ubiquitous network access is critical for IoT but often difficult to guarantee in remote or infrastructure – limited areas .
- Cost : The expensive involved in designing ,deploying and maintaining IoT networks can be high ,especially for smaller organizations .

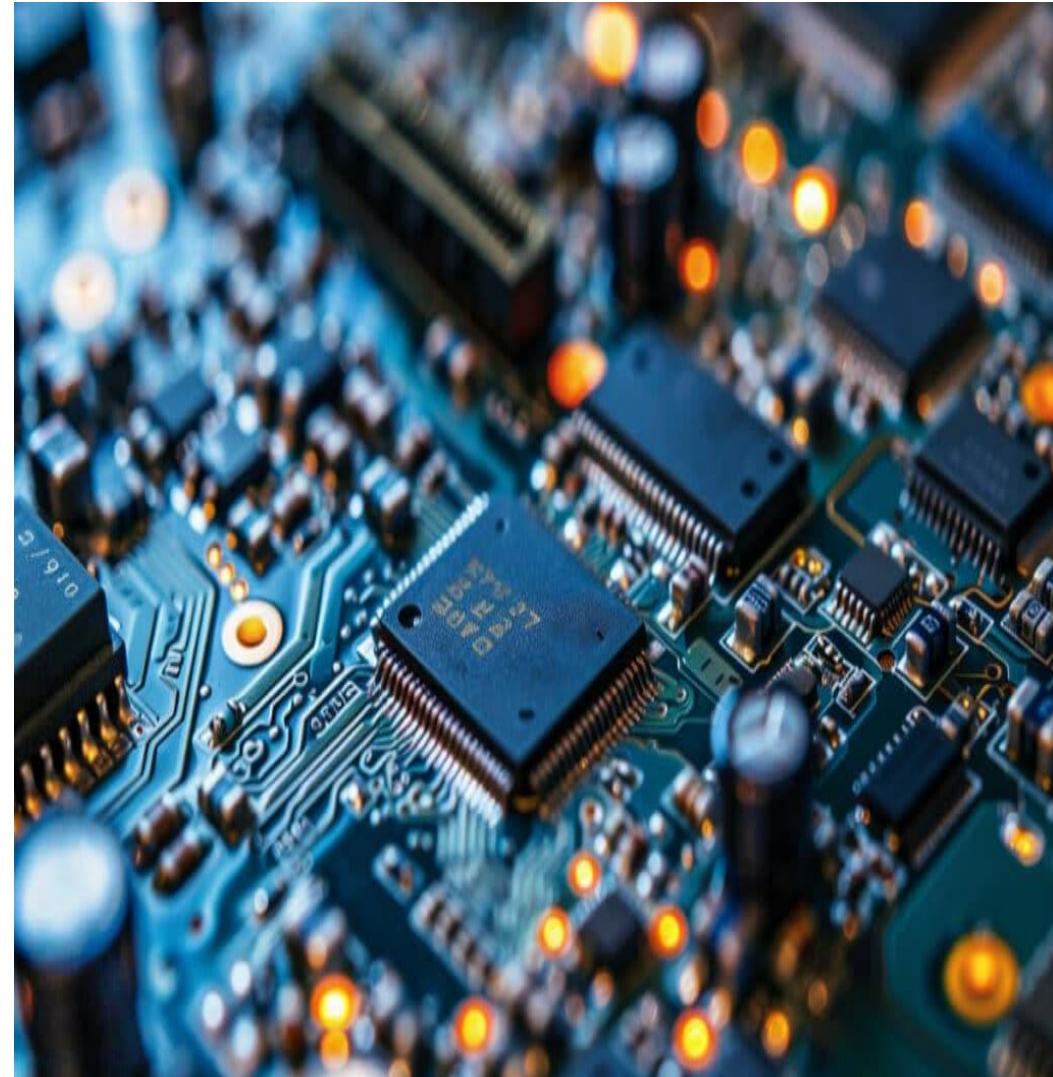


SOLUTION RELATED TO ISSUES FACED IN IOT

- Implement robust security protocols , such as encryption and strong authentication mechanisms ,to safeguard device data and prevent unauthorized access .
- Update software and firmware regularly to fix vulnerabilities and ensure devices are protected against evolving threats .
- Use zero-trust segmentation and network policies to limit access between devices , contain threats , and minimize attack surfaces .
- Deploy advanced threat detection technologies , including AI and machine learning systems , to monitor networks for suspicious behavior and automate responses .
- Standardize devices interoperability through accepted communication protocols and well-defined APIs , addressing compatibility issues between devices .
- Assign unique , cryptographically based identities to devices to track and authenticate each unit throughout its lifecycle .

EMBEDDED SYSTEM

An Embedded system is a specialized computing system designed to perform dedicated tasks of control specific functions within a larger device or applications. It typically consists of a microcontroller or microprocessor , memory components(RAM , ROM or flash) , and input/output interfaces for interfacing with sensors , actuators , or user controls . Embedded systems may include supporting software such as real-time operating systems(RTOS)and application programs , enabling them to execute instructions and react to environmental changes. These systems are optimized for efficiency , reliability , and minimal size and are widely used in products like smartphones , household appliances , medical devices , industrial machines , vehicles , and automation systems. Unlike general purpose computers , embedded systems focus on performing specific operations , often with real-time constraints , low power consumptions , and cost-effective hardware tailored for their target tasks.



PROJECT REQUIREMENTS

- Microcontroller : Arduino UNO
- Modules implemented: garden light control , temperature regulation , and serial tank management.
- Displays and Shields: CLCD and ethernet shield
- Simulation tools: used to develop and test each prototype before physical execution

SIMULATION TOOLS FOR PROJECT IMPLEMENTATION

- Arduino IDE : Platform for writing code and compiling it into HEX files for deployment .
- PICSIM LAB : virtual environment for simulating hardware and executing the compile files .
- NULL Emulator : Blynk IoT APP ,helping control systems like temperature and tank management remotely .

IMPLEMENTATION

- A folder is created in the name of the project including files like home_automation_Blynk_controlled . Ino . LDR . CPP , IDL. h, serial_tank . CPP , serial_tank . H , temperature_system . CPP , temperature_system . h , main . h .
- After coding in these files in Arduino IDE , export compiled binary file of the . INO file .
- In PICSIMLAB arrange parts like LED, LDR, LCD , temperature system, ethernet .
- Load hex file for Arduino UNO board in PICSIMLAB to view output.
- Use Blynk app in mobile to visualize and control the system.

ARDUINO UNO

The Arduino UNO is a widely used microcontroller board based on the ATmega328p chip, which features 32kb of flash memory, 2kb of SRAM, and 1kb of EEPROM for program and data storage. It operates at a clock speed of 60MHz and uses a 5V operating voltage. The board includes 14 digital input/output pins, of which 6 can be used as PWM outputs, and 6 Analog input pins labelled A0 through A5. Communication interfaces supported by the Arduino UNO include UART for serial communication SPI and I2C, allowing it to connect with a wide range of peripherals and sensors. The board is equipped with a USB port for programming and serial communication, and a power jack for external power sources, an ICSP header for in-circuit programming, and a reset button for restating the board. Its compact size (approximately 68.6mm*53.4mm) and user-friendly design make it highly suitable for beginners and professionals building embedded systems, automation projects, and IoT applications. The onboard LED connected to digital pin 13 is commonly used for testing and debugging purposes. The Arduino UNO is a cost-effective and reliable platform supported by a vast community and extensive libraries, making it an ideal choice for rapid proto typing and educational purposes.

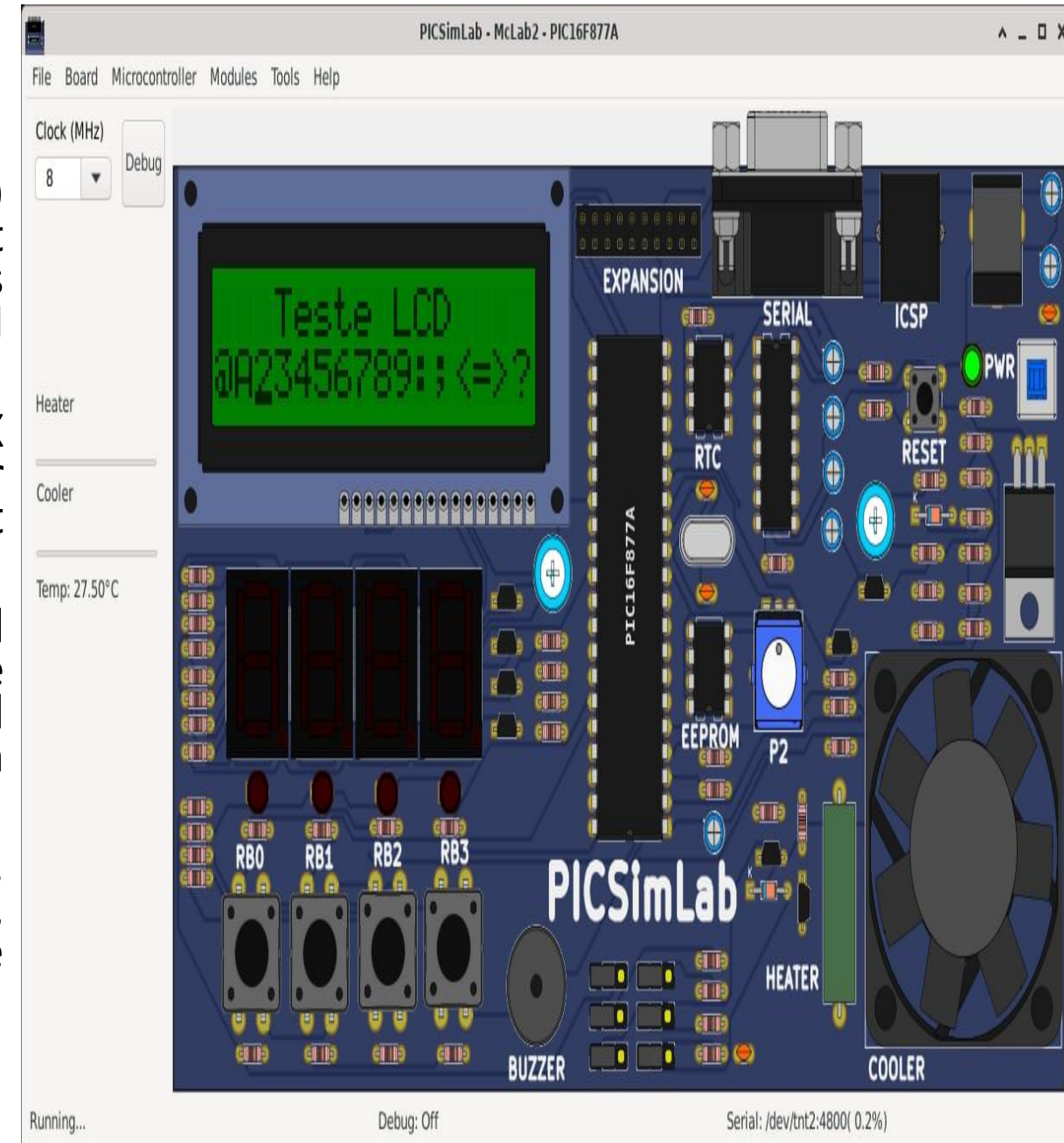


ARDUINO IDE SOFTWARE

- Arduino IDE (Integrated Development Environment) is a software platform used to write , compile and upload code to Arduino boards .
- It provides a simple interface for coding in a language similar to C/C++ designed for Arduino hardware interaction .
- Features include cross-platform support(Windows , macOS ,Linux) ,built-in code editor with syntax highlighting and auto-formatting .
- Includes a compiler that converts code into machine-readable instructions for Arduino microcontrollers .
- Supports one-click upload of code to Arduino boards via USB .
- Library manager facilitates adding custom libraries easily .
- Serial monitor is available to debug and view data sent from the Arduino board .
- The interface has components like a code editor , toolbar(for verify , upload ,new/open/save file , serial monitor) ,message area(for compile/upload status) , text console , and board/port selectors .

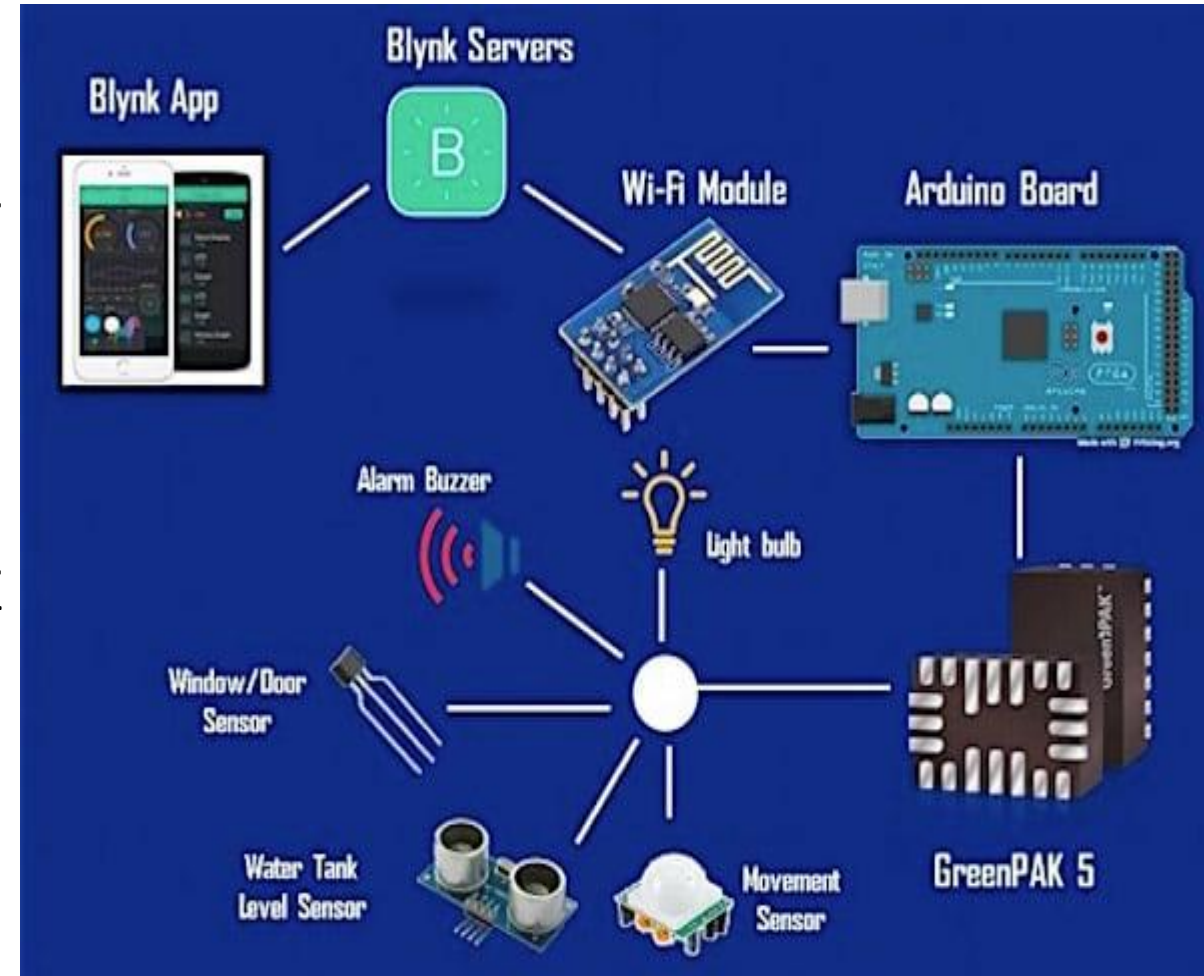
PICSIMLAB

- PICSIMLAB (Programmable IC Simulator Laboratory) is a real-time emulator for development boards that supports microcontrollers from various simulators like PICSIM , SIMAVR , uCsim,qemu-esp32 and GPISM .
- It integrates with development tools such as MPLABX , Arduino IDE and VSCODE with PLATFORMIO for programming , debugging and uploading code , but does not provide its own source code editor .
- PICSIMLAB simulates real hardware boards and allows connection of multiple devices (called spare parts) like LEDs , push buttons ,ethernet shield w5500 , and color graphic displays with touchscreen for comprehensive simulation .
- It aims to emulate real hardware behavior closely , enhancing debugging and testing flexibility , including supporting of microcontrollers like PIC16F628 and Arduino UNO with ATMEGA328P



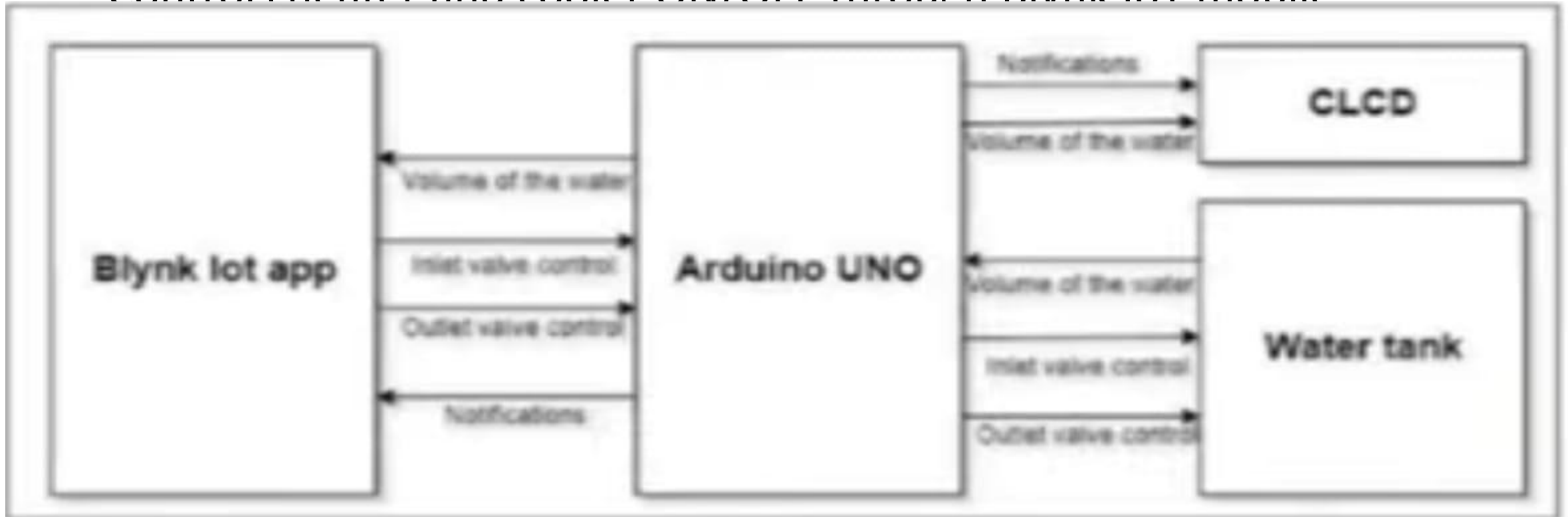
BLYNK

- The Blynk app is part of a comprehensive low-code IoT platform trusted by over 1 million developers worldwide . It enables users to build and customize feature-rich IoT mobile apps without writing code . Blynk simplifies the entire IoT process , including device activation , WIFI provisioning , remote control , sensor data visualization , OTA firmware updates , and secure cloud connectivity . It supports over 400 hardware boards (such as ESP32, Arduino , Raspberry Pi) and various connection methods including WIFI , ethernet , cellular ,LORAWAN ,HTTPS and MQIT .
- The platform consists of :
- Blynk . Apps: A drag-and-drop mobile app builder for IOS and android to monitor , manage , and control IoT devices remotely .
- Blynk . Console: A web portal for managing devices , users , organizations , OTA updates and business functions .
- Blynk . Cloud : Secure cloud infrastructure for hosting , scaling , and monitoring IoT solutions , with real-time data processing and API connectivity .



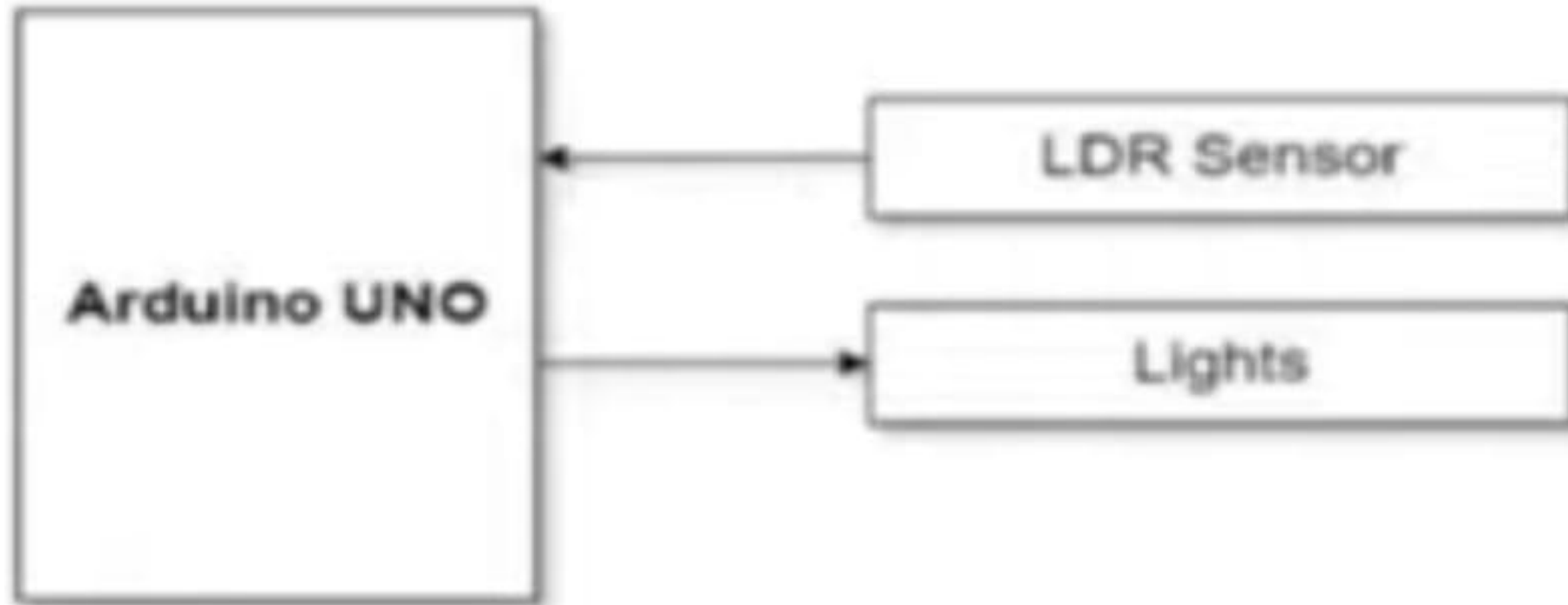
TEMPERATURE CONTROL SYSTEM

- Consists of heating resistor , LM35 temperature sensor , and cooler , resembling a home temperature control system .
- Reads temperature and displays it on CLCD .
- Controls heater and cooler ON/OFF through Blynk IoT mobile



GARDEN LIGHT CONTROL SYSTEM

- LDR Sensor is used Light Dependency Resistor .
- Resistance changes with intensity of light .
- Sensor detects changes in resistance to control garden li



WATER TANK CONTROL SYSTEM

- Reads volume of water in the tank through Serial Communication and displays in CLCD .
- Controls water volume by controlling inlet and outlet valves via serial commands .
- Notifications and volume display on Blynk mobile app .

CONCLUSION

- Using Blynk IoT application and PICSIMLAB simulator home automation where LED , temperature system , serial tank resemble light , heater , cooler and water tank in real time .
- CLCD acts as dashboard : Blynk app widgets control heater , cooler , inlet and outlet valves . IoT involves increasing numbers of smart interconnected devices and sensors used in home automation .
- IoT brings technological changes making life simpler and more comfortable .